Description of the contents of the data packages for the total solar eclipse of 12 August 2026 and 2 August 2027, and solar annular eclipse of 26 January 2028

The data package consists of 3 files per eclipse. The data provided are the same as in the eclipse viewers at **https://visualizadores.ign.es/eclipses**. Their use is subject to the conditions specified in the **IGN/CNIG Download Centre**, which are set out in a **CC-BY 4.0** licence. In all cases, it must be stated that this is "**Official astronomical information provided by the National Geographic Institute based on calculations made by the National Astronomical Observatory"**. For further details on the method used to perform the calculations, visit https://eclipses.ign.es/condiciones-de-observacion.html.

- **10bands_yyyy_3857_COG.tiff** (where yyyy indicates the year of the eclipse)
  10-band GeoTIFF file containing the eclipse ephemerids calculated from the geographical coordinates (latitude, longitude and elevation) of the GMTED2010 global digital elevation model (USGS and NGA) with a resolution of 30 arc sec. (approximately 800 m), in EPSG:3857 projection – WGS 84 / Pseudo-Mercator. The content of each of the bands is:
  band 1: elevation of the centre of the Sun in degrees, corrected for refraction, at the moment of maximum eclipse
  band 2: azimuth of the centre of the Sun in degrees at the moment of maximum eclipse
  band 3: obscuration at maximum
  band 4: sunrise on the day of the eclipse in UTC hours
  band 5: sunset on the day of the eclipse in UTC hours
  band 6: start of the partial eclipse in UTC hours
  band 7: moment of maximum eclipse in UTC hours
  band 8: end of the partial eclipse in UTC hours
  band 9: start of the totality / annularity phase (if applicable) in UTC hours
  band 10: end of the totality / annularity phase (if applicable) in UTC hours

- **terrain_shadows_yyyy_3857_COG_DEF_OVERSx6_8704x8192.tif** (yyyy is the year)
  A single band GeoTIFF file containing the terrain shadows at the moment of the eclipse's maximum. For both the calculation of the eclipse ephemeris (elevation and azimuth of the Sun at maximum) and the terrain, the GMTED2010 global digital elevation model (USGS and NGA) with a resolution of 7.5 arc seconds (approximately 200 m) was used, in EPSG:3857 projection – WGS 84 / Pseudo-Mercator. The file has a resolution of approximately 300 m. The 'insol' routine developed by Javier González Corripio was used to calculate the terrain shadows. Neither buildings nor trees were taken into account in this calculation; only the terrain relief was considered. On the map, a value of 0 indicates that the terrain fully blocks the view of the Sun at the moment of maximum obscuration; a value of 1 indicates that the terrain allows only part of the upper half of the Sun to be seen; a value of 2 indicates that the terrain allows the upper half of the Sun and part of the lower half to be seen; and a value of 3 indicates that the terrain does not block the view of the entire solar disc.

- **eclipse_levels_XXXX.gpkg** (where yyyy indicates the year of the eclipse)
  File in GeoPackage format containing the following groups of contour lines

  durtot_standard_yyyy.geojson (where yyyy indicates the year): group of contour lines showing the duration of the totality phase in seconds in GeoJSON format, at 10-second intervals, calculated from bands 9 and 10 in 10bands_yyyy_3857_COG.tiff, taking into account sunset (band 5). The resolution and projection are the same.

  oscurez_standard_yyyy.geojson (where yyyy indicates the year: group of contour lines showing maximum obscuration in GeoJSON format, at 10% intervals, calculated

from band 3 in 10bands_yyyy_3857_COG.tiff, taking into account sunset. The resolution and projection are the same.

moon_shadow_yyyy_merged_ff.fff-ll.lll-yyyy-simplified-0.0001.geojson (yyyy is the year, ff.fff is the UTC hour of the first curve, and ll.lll is the UTC hour of the last curve): collection of curves of the Moon's umbra /antumbra from ff:fff UTC (start of totality / annularity in the studied area) to ff:fff 567 UTC (end of totality / annularity in the studied area), at intervals of 0.001 hours (3.6 seconds). These curves are calculated from the data in bands 9 and 10 of the file 10bands_yyyy_3857_COG.tiff, taking into account the sunset (band 5).